THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM THREE FIRST CONTINUOUS ASSESSMENT TEST

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the assessment room.
6. Write your **Name in Full** on every page of your answer booklet(s).

KINDOROKO BIO 1 CAT 1 2025

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Which of the following is the correct classification of mushroom?

A Kingdom plantae; phylum bryophyta
B Kingdom fungi; phylum zygomycota
C Kingdom plantae; phylum pteridophyta
D Kingdom fungi; phylum basidiomycota
E Kingdom plantae; phylum ciliophora

(ii) Ashura is a botanist. This is to say that she is: -

A Specialized in the study of living things
B Specialized in the study of different plants
C Specialized in the study of animals like goats
D Specialized in the study of microorganisms
E An expert in scientific field, particulary biology

(iii) A boy has just started developing beards on his chin; hair under the armpits and producing sperms. What does this imply?

A He has reached the age of puberty
B He lacks enough proteins in the body
C He is developing marasmus
D He is not doing enough physical exercises
E He has a lot of fat in the body

(iv) Which of the following depends on cilia in order to change their position?

A Amoeba and euglena
B Paramecium and bacteria
C Ovum and paramecium
D Euglena and bacteria
E White blood cell and amoeba

(v) The basic structure used by spider for gaseous exchange is: -

A Tracheal system
B Gills
C Alveoli (*Sing: Alveolus*)
D Skin and lungs
E Book lungs

(vi) The main reason for why cypress trees produce soft timber is that:

A They do not produce flowers or fruits
B They form ever-green forests in tropical areas
C They possess needle-like leaves
D They reproduce by means of naked seeds
E They contain simple xylem and phloem

(vii) Faudhia was working in the laboratory. By accident she spilled unknown chemical on the floor which resulted into a permanent trace. Which of the following could be the most possible chemical that was unknown?

A Liquified petroleum gas
B Concentrated nitric acid
C Potassium permanganate solution
D Gunpowder or dynamite
E Dilute sodium hydroxide solution

(viii) Tics are commonly found on the skin of cattle such as cows and suck blood from them. This means that:

A Ticks are preys and cattle are predators
B Ticks are ectoparasites and cattle are hosts
C Ticks are ectoparasites and cattle are endoparasites
D Ticks are heterotrophs and cattle are autotrophs
E Ticks are blood suckers and cattle are primary producers

(ix) Radhmina and Bertha want to conduct an experiment to understand why the contents of an egg become solid when the egg is boiled. At this stage, which step would you recommend for them to begin with?

A Identification of the problem
B Experimentation
C Observation and data recording
D Hypothesis formulation
E Interpretation of data

(x) In the first aid procedure, which of the following items is used for removing thorns and stings left behind by bees?

A Tweezer
B Liniment
C Gentian violet
D Antiseptic
E Medical tapes

2. Match the items in **List A** with the responses in **List B** by writing the letter of the correct response besides the item number in the answer booklet(s) provided.

LIST A	**LIST B**
(i) Used to conduct experiments that emit harmful gases or smoke	A Benedict's solution
(ii) Used to test for the presence of starch in a given food sample	B Light telescope
(iii) Used to observe small organisms	C Fume chamber
(iv) Used to catch flying specimens like insects	D Iodine solution
(v) Used to store test tubes when they are not in use	E Hand lens
(vi) A small container with tools and items used to dissect specimens	F Sweep net
	G Test tube rack
	H Dissecting kit
	I Test tube holder
	J Fish net

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) What is a cell?
   (b) Outline the differences between mango cell and rod-shaped bacteria.
   (c) Draw the diagram of two adjacent guard cells to show their appearance at night.

4. (a) Give the reason(s) to justify each of the following facts:
      (i) Humans are said to be monogastric animals.
      (ii) Plants with yellowish leaves normally give poor yield.
      (iii) Salivary amylase enzyme cannot digest food in the stomach.
   (b) A group of students were arguing that scientists should find a means or eradicating all bacteria from our environment. If you are asked to oppose their idea, which points would you give out?

5. Carefully study the specimen below and then answer the questions that follow:



   (a) Name the kingdom, phylum/division and class to which the specimen above belongs.
   (b) Why have you placed the specimen above in the named class?
   (c) State three other characteristics of the class that you have described above.
   (d) The phylum/division to which the specimen above belongs is very advantageous to human being in various ways. Justify this statement by using two points.

6. (a) Distinguish between blood transfusion and blood compatibility.
   (b) Explain the functions of the main components of blood in the mammalian body.

7. The following organisms were found in the ecosystem: Grass, Toad, Vulture, Grasshopper, Lizard and Beetle.
   (a) Construct a food web from the organisms named above.
   (b) From the food web that you have constructed in (a), which organism(s) are:
      (i) Producers
      (ii) Secondary consumers
      (iii) In the second trophic level?
   (c) (i) Which important group of organisms should be introduced in the ecosystem above to make it balanced?
      (ii) Organisms of the group named in c(i) are commonly found into two kingdoms. Mention the kingdoms.

8. (a) Differentiate between endoskeleton and exoskeleton basing on: Position in the body, Composition and Examples of organisms which possess them.
   (b) It is common to see organisms moving from one place to another. How is this process important to the living things? Briefly explain by giving four points.
   (c) Briefly describe two adaptive features of the scapula to its functions.

SECTION C (30 Marks)
Answer **two (2)** questions in this section

9. There is a lot of microorganisms around us which can cause harm to our health. With six points, justify how our bodies defend themselves against such microorganisms.

10. Students at Kindoroko Girls' Secondary School are planning to stop cleaning the school environment for six months since they want to concentrate on studies only. Explain to them any six consequences of their decision.

11. Explain three types and three significances of transpiration in plants.

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